

PLA/PET Recycling Lab v0.8 — 최종 BOM

Revision final-design-fabrication-closure-v0.8 · 디지털 제조 후보

구매·가공·통전 승인 문서가 아니다. donor와 supplier가 미확정인 항목을 0원 또는 승인품으로 간주하지 않는다. 실제 라벨·치수·정격·수령검사와 사용자 승인이 필요하다.

1 구성관리 기준

- Authoritative design source: bom/bom.csv (SHA-256 58db3a9ad75b9d1cf396f2091368b6a6151dbc273caf90519d0bd4afbceb0427)
- Subordinate detail sources: print, machine, shredder, extruder, thermal, drive manifests
- BOM rows: 110 / source design rows: 32
- 공급 상태 집계: CANDIDATE_SELECTED—RECEIPT_TEST_REQUIRED: 2, DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED: 54, DONOR_IDENTIFICATION_REQUIRED: 1, GATE1_QTY_1_ALLOWED_AFTER_DONOR_MEASUREMENT_AND_USER_APPROVAL: 1, NOT_APPLICABLE_REFERENCE: 6, PRINT_PACKAGE_PASS: 12, REFERENCE_ONLY_NOT_SELECTED: 1, RFQ_NOT_SENT—USER_APPROVAL_REQUIRED: 2, SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED: 10, SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED: 7, USER_APPROVAL_AND_RECEIPT_TEST_HOLD: 4, USER_INVENTORY_VERIFICATION_REQUIRED: 4, USER_SELECTION_OR_APPROVAL_REQUIRED: 6
- Price는 정보이며 설계 release gate가 아니다. 미확정 MPN/공급사는 주문 전에 달는다.
- CSV의 17개 필드가 전체 release record이며 이 PDF는 사람이 검토하는 vector view다.

2 조달·제작 보기

Part ID	설명	수량	Make/Buy	재료/사양	승인 MPN	공급 상태
CO-01	vertical compact air cooling	1	MIXED	ABS duct and donor fans	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
CO-02	cooling fan current plus independent tach feedback	1	MIXED	calibrated A4 current feedback plus two fan tach lines through 2:1 mux to A14	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
CT-01	Arduino Mega user interface	1	VERIFY_REUSE_OR_BUY	CT-ENC-01 sheet enclosure plus Mega TFT encoder buttons and fixed 18x18 grounded cable ducts	NONE_APPROVED—exact make/model and receipt evidence pending	USER_INVENTORY_VERIFICATION_REQUIRED
CT-ENC-01	Control-panel sheet enclosure	1	MAKE_TO_DRAWING	2 mm 5052 aluminum; process=laser + brake + PE stud	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-01	Cycloidal hook cutter disc	12	MAKE_TO_DRAWING	6 mm AISI D2 tool steel (JIS SKD11 equivalent); process=rough waterjet, vacuum harden/double temper 58-60 HRC, finish profile/faces grind	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-02	Cutter spacer	10	MAKE_TO_DRAWING	S45C normalized steel; process=simple turning	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-03	Bearing side plate	2	MAKE_TO_DRAWING	12 mm S275JR steel; process=laser + bearing-seat finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-04	5 mm aperture screen	2	MAKE_TO_DRAWING	3 mm 304 stainless; process=laser cut + deburr	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-05	20 mm keyed cutter shaft	2	MAKE_TO_DRAWING	S45C normalized steel HB 170-229; process=turn + keyway; black oxide with bearing seats masked	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-06	Phase gear axial spacer	2	MAKE_TO_DRAWING	S45C normalized steel; process=simple turning	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-07	DRV-01 universal donor motor plate	1	MAKE_TO_DRAWING	6 mm steel; process=laser cut + deburr; standard metal angles	NOT_APPLICABLE—make from released file	GATE1_QTY_1_ALLOWED_AFTER_DONOR_MEASUREMENT_AND_USER_APPROVAL
CUT-08	Dual 6004 bearing retainer	2	MAKE_TO_DRAWING	2 mm steel; process=laser cut + deburr	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DG-01	dual-axis shadow gauge	1	BUY	LED photodiode slit	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
DR-EXT	external qualified pre-dryer	1	USER_SUPPLIED_EXTERNAL	project lab/user equipment	NOT_APPLICABLE—build to released specification	USER_INVENTORY_VERIFICATION_REQUIRED
DRV-01	Universal donor motor plate	1	MAKE_TO_DRAWING	6 mm SS400 steel; process=laser cut + standard metal angles	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-02	Bolt-on cutter sprocket hub	1	MAKE_TO_DRAWING	S45C; process=turn + keyway + PCD drilling	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-03	M3 Z16 keyed phase gear lamination	6	MAKE_TO_DRAWING	6 mm S45C; process=waterjet/wire-EDM profile + stack dowel/finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-A42	42GP-775 reference adapter plate	1	MAKE_TO_DRAWING	6 mm SS400 steel; process=laser + drill/ream	NOT_APPLICABLE—make from released file	REFERENCE_ONLY_NOT_SELECTED
DRV-A60	GMP60-60127 selected adapter plate	1	MAKE_TO_DRAWING	6 mm SS400 steel; process=laser + drill/ream	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-F01A	Motor-side fuse inner hub	1	MAKE_TO_DRAWING	S45C; process=turn + keyway + radial drill	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-F01B	Motor-side fuse sprocket carrier	1	MAKE_TO_DRAWING	S45C; process=turn + PCD drill + radial drill	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-F01P	Replaceable waisted shear pin coupon	6	MAKE_TO_DRAWING	C360/CuZn39Pb3 brass; process=turn waist to released digital baseline; optional coupon correlation may refine later	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-GD-01	Interlocked drive guard	1	MAKE_TO_DRAWING	1 mm galvanized steel; process=laser + brake + service-cover hardware	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-01	16 mm x 16D screw barrel and EX-DIE-01 to 05 connected open-die	1	MAKE_CNC	SCM440/304/17-4PH/C110	NOT_APPLICABLE—build to released specification	RFQ_NOT_SENT—USER_APPROVAL_REQUIRED
EX-02	thrust bearing plate clamps	1	MAKE_CNC	EX-THR-01 metal	NOT_APPLICABLE—build to released specification	RFQ_NOT_SENT—USER_APPROVAL_REQUIRED
EX-03	barrel process heater zones	3	BUY_CUSTOM	TH-BH-01 custom 24 V 100 W ID34 W45 mica band	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
EX-04	die heater	1	BUY	TH-DIE-01 24 V 60 W Ø6x38 cartridge	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
EX-05	thermal sensors and switching	1	MIXED	T1-T5 ungrounded mineral-insulated K-type plus MAX6675-class common-reference interfaces and 5 low-side MOSFET channels	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
EX-06	screw shaft tach	1	MIXED	Hall sensor plus one or more shaft magnets and protected A13 input	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
EX-BAR-01	ID16.20 x OD34 barrel	1	MAKE_TO_DRAWING	SCM440 (KS D3867/JIS G4105 equivalent) QT + gas nitride bore; process=deep drill, stress relieve, ream/hone, port, die-interface thread and sensor bores, nitride, final hone	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-CPN-BAR	Matched barrel process coupon	1	MAKE_TO_DRAWING	same certified SCM440 heat as EX-BAR-01; process=same bore/hone/nitride route as EX-BAR-01	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-CPN-SCR	Three-pitch screw process coupon	1	MAKE_TO_DRAWING	same certified SCM440 heat as EX-SCR-01; process=same flight mill/polish/nitride route as EX-SCR-01	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-01	Connected 90 degree down-die body	1	MAKE_TO_DRAWING	SCM440 QT + gas nitride; process=6-face mill; gun drill/ream intersecting Ø8 channels; counterbore, drill/tap; stress relieve; gas nitride; lap sealing face	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-02	Seven-hole breaker plate	1	MAKE_TO_DRAWING	304 stainless; process=wire EDM or laser + double-side lap	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED

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EX-DIE-03	Replaceable Ø3 die insert	1	MAKE_TO_DRAWING	17-4PH H900 stainless; process=turn, drill/ream land, 60 degree included entrance blend, H900, finish lap	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-04	Sacrificial die relief retainer	1	MAKE_TO_DRAWING	304 stainless sheet t1.5; process=laser/waterjet + deburr; no heat treatment	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-05	Annealed copper face gasket	2	MAKE_TO_DRAWING	C110 annealed copper t0.5; process=waterjet/punch; anneal after cutting; bag clean	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-MT-01	rear hot-zone axial datum plate	1	MAKE_CNC	S275 steel plate t8	NOT_APPLICABLE—build to released specification	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
EX-MT-02	front hot-zone radial sliding guide	1	MAKE_CNC	S275 steel plate t8	NOT_APPLICABLE—build to released specification	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
EX-MT-03	rear split fixed collar	1	MAKE_CNC	S45C	NOT_APPLICABLE—build to released specification	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
EX-MT-04	rear hot-zone support rail	1	MIXED	2020 aluminum extrusion L390	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
EX-SCR-01	16 mm x 16D single screw	1	MAKE_TO_DRAWING	SCM440 (KS D3867/JIS G4105 equivalent) QT + gas nitride; process=turn between centres, 4-axis flight mill, polish, nitride, finish grind	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-SH-01	Three-panel hot-zone shield	1	MAKE_TO_DRAWING	2 mm 5052 aluminum; process=laser + two 90deg bends; bond PE	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-THR-01	Extruder thrust plate	1	MAKE_TO_DRAWING	12 mm S45C normalized steel; process=laser rough + bore/seat finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
ExtruderFixed Collar	reference alias: EX-MT-03	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
ExtruderFront SlidingGuide	reference alias: EX-MT-02	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
ExtruderRear FixedDatum	reference alias: EX-MT-01	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
Extruder SupportRail Rear	reference alias: EX-MT-04	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FB-01	removable flake bin	1	MIXED	FD-BIN-01 1 mm sheet plus 4x PPR-C03	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
FD-BIN-01	Removable flake bin	1	MAKE_TO_DRAWING	1 mm 304 stainless sheet; process=laser cut, fold and rivet/TIG seam	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-HOP-01	Sealed feed hopper	1	MAKE_TO_DRAWING	2 mm 304 stainless; process=roll cone/cylinder + TIG weld + gasketed lid	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-MET-01	Metering auger housing	1	MAKE_TO_DRAWING	304 stainless; process=turn tube/flanges + drill	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-MET-02	Positive-displacement metering auger	1	MAKE_TO_DRAWING	304 stainless; process=turn root/bore + mill or weld continuous flight	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-MET-03	Common auger and anti-bridge agitator shaft	1	MAKE_TO_DRAWING	304 shaft; process=turn Ø8 shaft + weld/pin Ø4 paddles	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-TRN-01	Sealed transfer tube	1	MAKE_TO_DRAWING	304 tube OD32 x2; process=tube cut + socket fit + TIG tack/weld	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FH-01	sealed steep-wall hopper with coaxial anti-bridge agitator and positive metering auger	1	MIXED	FD-HOP-01 plus FD-MET-01..03 auger housing removable auger and Ø8 common agitator drive shaft	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
FH-02	hopper maintenance PTC assembly	1	MIXED	4x 35x21x5 class PTC plus TH-PTC-01/02 metal spreader clamp	NONE_APPROVED—exact make/model and receipt evidence pending	CANDIDATE_SELECTED—RECEIPT_TEST_REQUIRED
FH-03	single metering-rotor drive sensing interface	1	MIXED	one donor geared motor plus driver fault and low-speed tach feedback	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
FM-AX-01	Puller roller spindle	2	MAKE_TO_DRAWING	Ø8 h6 stainless shaft; process=cut/face + collar flats	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-GA-01	Guide roller fixed axle	1	MAKE_TO_DRAWING	Ø5 h6 stainless shaft; process=cut/face + E-clip grooves or collars	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-GR-01	Solid-strand guide roller	1	MAKE_TO_DRAWING	POM-C; process=turn + bearing-seat bore	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-PL-01	Puller side plate	2	MAKE_TO_DRAWING	10 mm 6061-T6; process=waterjet + ream	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-RL-01	Puller roller	2	MAKE_TO_DRAWING	6061-T6 hub + replaceable Shore A 50-70 silicone sleeve; process=turn + bore	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FR-01	compact profile frame cut set	1	VERIFY_REUSE_OR_BUY	20x20 aluminum profile 890x4 430x10 660x6 300x2 318x1 280x2 50x1 plus local 20x40 660x2	NONE_APPROVED—exact make/model and receipt evidence pending	USER_INVENTORY_VERIFICATION_REQUIRED
GD-01	cutter and hot-zone guards	1	MIXED	metal sheet plus ABS cold covers	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
HP-01	refillable sliding-lid hopper	1	MIXED	IN-HOP-01 plus PPR-C01/C02	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
IN-HOP-01	Refillable input hopper	1	MAKE_TO_DRAWING	2 mm 5052-H32 aluminum; process=roll cone/cylinder + TIG weld + deburr	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
PL-01	puller pair with tach	1	MIXED	FM-PL-01/RL-01/AX-01 plus donor motor and tach	NONE_APPROVED—exact make/model and receipt evidence pending	USER_INVENTORY_VERIFICATION_REQUIRED
PPR-C01	Sliding hopper lid	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.24 mm; 4 walls; 20% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C02	Anti-reach baffle chute	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.24 mm; 5 walls; 25% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C03	Flake bin sheet corner	4	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.28 mm; 4 walls; 25% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C04	Screen drawer handle	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.24 mm; 5 walls; 35% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C05	Cooling duct segment	2	MAKE_3D_PRINT	ABS; process=FDM; 0.4 mm nozzle; 0.24 mm; 4 walls; 15% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C06	Gauge enclosure half	2	MAKE_3D_PRINT	ABS; process=FDM; 0.4 mm nozzle; 0.20 mm; 4 walls; 25% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C07	Puller pinch guard	1	MAKE_3D_PRINT	ABS; process=FDM; 0.4 mm nozzle; 0.24 mm; 5 walls; 20% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C08	Solid-strand guide axle bracket	2	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 5 walls; 40% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C09	Spool cone adapter	2	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 5 walls; 35% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C10	Traverse carriage	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 5 walls; 40% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C11	Control panel bezel	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 4 walls; 20% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C12	Cable duct clamp	8	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 4 walls; 50% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-EXTRUDER-ASM	reference alias: PPR-EXTRUDER-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-FEEDER-ASM	reference alias: PPR-FEEDER-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-FORMING-ASM	reference alias: PPR-FORMING-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE

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PPR-FRAME-ASM	reference alias: PPR-FRAME-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-FULL-ASM	reference alias: PPR-FULL-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-SHREDDER-ASM	reference alias: PPR-SHREDDER-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
SF-01	latching E-stop and hardwired cut	1	BUY	qualified switch plus relay topology	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SF-02	lid and service interlocks	3	BUY	positive opening switch candidate	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SF-03	thermal and branch fuses	1	BUY	temperature/current rated	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SH-01	dual-shaft hook cutter assembly	1	MAKE_CNC	steel	NOT_APPLICABLE—build to released specification	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SH-02	removable 5 mm screen	1	MAKE_CNC	stainless steel	NOT_APPLICABLE—build to released specification	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SH-03	interchangeable shredder geared DC drive	1	VERIFY_REUSE_OR_BUY	GMP60-60127 ratio47 digital reference or accepted 18-30 V donor plus DRV-01/Axx	NONE_APPROVED—exact make/model and receipt evidence pending	DONOR_IDENTIFICATION_REQUIRED
SH-04	interchangeable chain and phase drive	1	MIXED	DRV-Axx/DRV-F01 plus #35 12T to 18/24/30T cutter-side DRV-02 and M3 Z16 face18 phase pair	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SH-05	bidirectional motor driver/current feedback	1	BUY	BTS7960-class plus 50 A Hall sensor	NONE_APPROVED—exact make/model and receipt evidence pending	CANDIDATE_SELECTED—RECEIPT_TEST_REQUIRED
SP-01	dancer traverse spooler	1	MIXED	FM-GR/GA SP-DA/AX/RL/SH plus donor motion tach left/right limits and PPR-C08..10	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
SP-AX-01	Dancer pivot/roller axles	2	MAKE_TO_DRAWING	Ø8 h6 stainless shaft; process=cut/face + collars	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-BP-01	Spool 6001 bearing plate	2	MAKE_TO_DRAWING	5 mm 6061-T6; process=waterjet + bearing-seat finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-DA-01	Dancer arm	1	MAKE_TO_DRAWING	8 mm 6061-T6; process=waterjet + ream	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-DS-01	Dancer pivot support plate	1	MAKE_TO_DRAWING	8 mm 6061-T6; process=waterjet + ream	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-MM-01	Universal NEMA17-class spool motor plate	1	MAKE_TO_DRAWING	6 mm 6061-T6; process=laser/waterjet + drill	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-RL-01	Dancer end roller	1	MAKE_TO_DRAWING	POM-C; process=turn + bore	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-SH-01	Spool spindle	1	MAKE_TO_DRAWING	Ø12 h6 S45C; process=cut/turn faces + collar flats	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-TR-01	Traverse rod end plate	2	MAKE_TO_DRAWING	5 mm 6061-T6; process=waterjet + ream	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
TH-BH-01	Custom barrel mica band heater	3	BUY_TO_SPEC	mica/NiCr/stainless sheath; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
TH-DIE-01	Die cartridge heater	1	BUY_TO_SPEC	stainless-sheathed cartridge; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	USER_APPROVAL_AND_RECEIPT_TEST_HOLD
TH-FUSE-01	Independent one-shot thermal fuse envelope	3	BUY_TO_SPEC	300 C-class barrel/die plus hopper-specific lower setpoint; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	USER_APPROVAL_AND_RECEIPT_TEST_HOLD
TH-PTC-01	Hopper PTC aluminum spreader	1	BUY_TO_SPEC	6061-T6 t3; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
TH-PTC-02	Hopper PTC grounded keeper	1	BUY_TO_SPEC	304 stainless t2; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
TH-PTC-EL	Hopper maintenance PTC element	4	BUY_TO_SPEC	24 V ceramic PTC; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	USER_APPROVAL_AND_RECEIPT_TEST_HOLD
TH-TC-01	Ungrounded K-type barrel/die/hopper probe	5	BUY_TO_SPEC	Ø3 mineral-insulated 304 sheath, ungrounded junction; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	USER_APPROVAL_AND_RECEIPT_TEST_HOLD

3 인터페이스·도면·조립 교차참조

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
CO-01	hot-shield clearance and verified airflow feedback	docs/drawings/v0.8/FM-001_cooling_strand_path.svg	docs/build_manual_ko.typ § Cooling, gauge, puller	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; two serviceable duct segments
CO-02	hot-shield clearance and verified airflow feedback	docs/drawings/v0.8/FM-001_cooling_strand_path.svg	docs/build_manual_ko.typ § Cooling, gauge, puller	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=USER_APPROVAL_REQUIRED; quantity_unit=each; commanded PWM with 1.5 s bounded feedback dwell; each fan stopped/both stopped/implausible-off detection; tach is not airflow
CT-01	PE bond, segregated routing and pin-map consistency	docs/drawings/v0.8/EL-001_electrical_enclosure.svg	docs/build_manual_ko.typ § Control과 UI	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=project lab; cash_class=REUSE_PENDING; status=UNVERIFIED; quantity_unit=each; profile lock and cleaning wizard; heater and sensor conductors require partitioned routing; duct stock receipt measurement pending
CT-ENC-01	PE bond, segregated routing and pin-map consistency; drawing tolerance=190 x35 x190; service-open face; PPR-C11 bezel datum; M4 profile mounts; segregate heater/motor and signal wiring; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CT-ENC-01.pdf	docs/build_manual_ko.typ § Control과 UI	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-01	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=OD 58.0; root circle R18.0 with smooth DXF transition; bore 20.2 +0.10/0; keyway width 6.2 +0.10/0; final profile ±0.10 and taper <=0.05 through thickness after CAM kerf compensation; t6.00 ±0.03; flatness/parallelism 0.03; tooth side deburr C0.15 max; axial working gap 0.25-0.50 mm by metal shim; common 6x6 key plus engraved index line registers stack; replace on crack, chip >0.5 mm, edge recession >0.30 mm or lost shim gap; datum=A=shaft bore axis; B=ground stack face; C=keyway centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-01.pdf	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-02	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=OD 28.0; bore 20.2 +0.10/0; length 7.00 +/-0.03; faces parallel within 0.03; datum=A=shaft bore axis; B=ground stack face; C=keyway centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-02.pdf	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-03	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=two 6004 seats diameter 42 H7; center distance 48.00 +/-0.03; match-machine both plates; seat-axis parallelism 0.05/140; four frame holes diameter 6.6; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-03.pdf	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-04	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=135 x 120 x 3; apertures diameter 5.0 on 9.0 pitch; all strand-side edges R0.3; verify minimum 1.9 mm rotating clearance with shims before powered test; datum=A=shaft bore axis; B=ground stack face; C=keyway centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-04.pdf	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-05	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=diameter 20 h6 at two 6004 journals per shaft; overall 240.0 +/-0.10; TIR <=0.05; 6 mm keyways at y=0-35, 55-160 and 195-240 from datum end; keyway depth 3.5; install driven shaft at Y258 and slave shaft at Y278 to preserve 20 mm front sprocket clearance; use standard metal clamp collars for axial retention; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-05.pdf	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-06	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=OD 30.0; bore 20.2 +0.10/0; length 4.00 +/-0.03; faces parallel within 0.03; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-06.pdf	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-07	shaft/bearing fit; cutter shim clearance; phase registration	exports/cnc/CUT-07/drawing_notes.md	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
CUT-08	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=figure-eight OD lobes 60; two relief bores diameter 34; center distance 48.00 +/-0.05; six M4 clearance holes diameter 4.5 at drawing coordinates; CUT-03 matching holes are included and may be match-reamed after bearing-seat finish; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-08.pdf	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DG-01	orthogonal gauge alignment and calibration	docs/drawings/v0.8/FM-002_gauge_puller.svg	docs/build_manual_ko.typ § Cooling, gauge, puller	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=MIXED; status=CALIBRATION_HOLD; quantity_unit=each; no accuracy claim
DR-EXT	external dryer qualification and material moisture evidence	docs/drawings/v0.8/GA-001_general_arrangement.svg	docs/build_manual_ko.typ § 작업 전 확인	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=external; cash_class=EXCLUDED; status=UNVERIFIED; quantity_unit=each; not inside machine envelope or cash budget
DRV-01	donor shaft interface; chain alignment; replaceable shear element; drawing tolerance=180 x140 x6; frame holes 4xØ6.6; slots nominal per DXF; flatness <=0.5; motor-specific adapter carries motor load; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-01.pdf	docs/build_manual_ko.typ § Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-02	donor shaft interface; chain alignment; replaceable shear element; drawing tolerance=bore Ø20.2 +0.03/0 after	exports/final/manufacturing/RFQ/DRV-02.pdf	docs/build_manual_ko.typ § Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
	received CUT-05 measurement; 6.2 keyway; 4xØ6.6 PCD36; sprocket register TIR <=0.10; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report				approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-03	donor shaft interface; chain alignment; replaceable shear element; drawing tolerance=M3 Z16 20 degree; bore Ø20.2 +0.05/0; internal 6.2 wide x6 radial keyway; 2xM4 clearance Ø4.5 + 1xØ3 H7 dowel on PCD30 at 0/120/240 degree; three laminations per gear; stack face >=18; key transmits shaft torque; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-03.pdf	docs/build_manual_ko.typ \$ Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-A42	donor shaft interface; chain alignment; replaceable shear element	exports/drive_interface/parts/DRV-A42/drawing_notes.md	docs/build_manual_ko.typ \$ Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
DRV-A60	donor shaft interface; chain alignment; replaceable shear element; drawing tolerance=80 x80 x6; centre Ø32.2; 4xØ5.5 PCD45; two 6.6x18 universal slots; match official motor drawing before order; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-A60.pdf	docs/build_manual_ko.typ \$ Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-F01A	donor shaft interface; chain alignment; replaceable shear element; drawing tolerance=Ø12.2 bore matched to received GMP60 Ø12 keyed shaft; OD20 x16; radial Ø3.2; no set-screw-only torque path; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-F01A.pdf	docs/build_manual_ko.typ \$ Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-F01B	donor shaft interface; chain alignment; replaceable shear element; drawing tolerance=OD36/ID20.4 x10; radial Ø3.2; 4xØ4.5 PCD28 to 12T sprocket; rotates free after pin fracture; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-F01B.pdf	docs/build_manual_ko.typ \$ Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-F01P	donor shaft interface; chain alignment; replaceable shear element; drawing tolerance=Ø3 x36 blank; starting waist Ø1.8 x4; target 10.35 N.m motor-side; one-shot BROKEN state; optional empirical calibration is not a design-release gate; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-F01P.pdf	docs/build_manual_ko.typ \$ Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-GD-01	donor shaft interface; chain alignment; replaceable shear element; drawing tolerance=165 x48 x190; two Ø26 shaft clearances at X20/68,Z55; open service face; positive-opening interlock flag; PE bond; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-GD-01.pdf	docs/build_manual_ko.typ \$ Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-01	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=RFQ_HOLD; quantity_unit=each; common PLA/PET hot path; physical process/hot-relief coupons before full order
EX-02	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=RFQ_HOLD; quantity_unit=each; screw to profile load path; Ø17.2 passage Ø30.2 seat
EX-03	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=custom RFQ; cash_class=CASH; status=USER_APPROVAL_HOLD; quantity_unit=each; 300 W total; stock Ø35 substitution prohibited; branch fuse and independent thermal fuse mandatory; vendor candidates pinned in bom/procurement_v0.5_ko.md (한국전 자전열 heater1.co.kr
EX-04	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=CASH; status=USER_APPROVAL_REQUIRED; quantity_unit=each; Ø6.05 H7 reamed bore; 0.070-0.122 mm diametral clearance; total process heat 360 W; candidate listing is not purchase approval; receipt test OD/camber/cold resistance 9.60 Ω ±10 percent at 20 C
EX-05	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=USER_APPROVAL_REQUIRED; quantity_unit=each; each channel branch fuse; sheath-to-junction insulation receipt test; MAX6675 T-common electronics reference; software is not sole cutoff; vendor-pinned modules are purchase candidates only
EX-06	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=USER_APPROVAL_REQUIRED; quantity_unit=each; actual RPM/revolutions gate purge; mass per revolution is not calibrated and purge mass is not claimed
EX-BAR-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=L280.00; ID16.20 +0.02/0 after hone; OD34.00 -0.03/0 after finish grind; 4x M4-6H depth8 PCD26; 3x Ø3.20 +0.05/0 blind5.5 sensor bores at B+95/170/245; minimum bore ligament 3.32; outer/inner thread ligament >=2.0/2.9; Datum B rear face/C front face; bore axis Datum D; SCM440 QT 28-32 HRC; gas nitride 0.30-0.50 mm, surface ≥900 HV; final hone bore Ra 0.4-0.8 μm; three-station bore report; datum=A=common journal/bore axis; B=drive/feed datum face; C=feed-port angular plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-BAR-01.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-CPN-BAR	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=L60.00; ID/OD/finish/case same as barrel; coupon RFQ only; datum=A=common journal/bore axis; B=drive/feed datum face; C=feed-port angular plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-CPN-BAR.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv

Part ID	핵심 인터페이스	도면	조립 단계	평웨어 의존	비고
EX-CPN-SCR	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=L48.00; three pitches; OD/root/land/finish/case same as feed zone; coupon RFQ only; datum=A=common journal/bore axis; B=drive/feed datum face; C=feed-port angular plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-CPN-SCR.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-DIE-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=40 x40 x48; barrel datum face X40; Ø8 melt turn; Ø16.20 +0.05/0 x3 breaker seat; Ø12.00 +0.03/0 x14 insert seat; 4x Ø4.5 + Ø8 head recess PCD26; heater Ø6.05 H7 reamed; sensor Ø3.20 blind12; face flatness 0.03; channel intersection fully deburred; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-DIE-01.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-DIE-02	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=Ø15.90 -0.05/0 x2.00 ±0.03; 7x Ø2.00 +0.05/0, six on PCD10; flatness 0.03; all flow edges R0.15 max; HOLD with die body; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-DIE-02.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-DIE-03	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=OD Ø11.90 -0.02/0 x14.00 ±0.03; outlet Ø3.00 +0.02/0 x10.00 land; 4 mm transition from Ø8 to Ø3; land Ra<=0.4 um; concentricity 0.02 to OD; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-DIE-03.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-DIE-04	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=32 x20 x1.5; two 10 wide x2.5 long bending webs; 2x Ø4.5 at 24 centres; centre bypass Ø4; flatness 0.15; coupon-calibrate at operating temperature, analytical estimate is not release evidence; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-DIE-04.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-DIE-05	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=OD34; ID16.20 +0.10/0; 4x Ø4.5 PCD26 at 45 degree; t0.50 ±0.03; burr <=0.03; one spare required; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-DIE-05.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-MT-01	rear axial datum/front sliding thermal expansion path	exports/final/manufacturing/hot_zone/ExtruderRearFixedDatum.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=DESIGNED; quantity_unit=each; v0.8 STEP/DXF/PDF; bore Ø34.10 +0.05/0; procurement USER_APPROVAL_REQUIRED
EX-MT-02	rear axial datum/front sliding thermal expansion path	exports/final/manufacturing/hot_zone/ExtruderFrontSlidingGuide.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=DESIGNED; quantity_unit=each; v0.8 STEP/DXF/PDF; bore Ø34.60 +0.10/0; cold axial travel at least 1.30 mm
EX-MT-03	rear axial datum/front sliding thermal expansion path	exports/final/manufacturing/hot_zone/ExtruderFixedCollar.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=DESIGNED; quantity_unit=each; v0.8 STEP/DXF/PDF; bore Ø34.10 +0.03/0; front guide installation prohibited
EX-MT-04	rear axial datum/front sliding thermal expansion path	exports/final/manufacturing/hot_zone/ExtruderSupportRailRear.svg	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/cut; cash_class=CASH; status=DESIGNED; quantity_unit=each; v0.8 STEP/DXF/PDF; metal face-joint; cut length 390.0 ±0.5 mm
EX-SCR-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=total 316.00; active 256.00; OD 15.92 -0.02/0; RH pitch 16.00; land 1.60; Datum A axis from Ø12 h6 and Ø15 h6 journals; SCM440 QT 28-32 HRC; gas nitride 0.30-0.50 mm, surface 900-1100 HV; flight OD TIR ≤0.05/256; OD Ra≤0.8 µm; datum=A=common journal/bore axis; B=drive/feed datum face; C=feed-port angular plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-SCR-01.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-SH-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=335 x75 x85; open bottom/ends; feeder opening Ø50 at X314/Y37; >=10 mm ABS-duct gap; edge hem/deburr; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-SH-01.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-THR-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=12 x95 x105; passage Ø17.2; thrust seat Ø30.2 x5; 4xØ6.6; seat axis square 0.05; metal-to-profile load path; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/EX-THR-01.pdf	docs/build_manual_ko.typ \$ Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
ExtruderFixed Collar	reference identity must resolve to canonical BOM/drawing; drawing tolerance=OD Ø50; L12.00 ±0.05; bore Ø34.10 +0.03/0; datum face runout 0.05 to bore axis; bore Ra≤1.6 µm; blue-fit contact ≥70%; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/ExtruderFixedCollar.pdf	docs/build_manual_ko.typ \$ 조립-체결 schedule	NONE	active_part_set reference; canonical=EX-MT-03; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
ExtruderFront SlidingGuide	reference identity must resolve to canonical BOM/drawing; drawing tolerance=guide bore Ø34.60 +0.10/0; 2×Ø6.6 rail holes; datum face flatness 0.10; bore position ±0.10; cold axial travel ≥1.30 mm; no axial clamp; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/ExtruderFrontSlidingGuide.pdf	docs/build_manual_ko.typ \$ 조립-체결 schedule	NONE	active_part_set reference; canonical=EX-MT-02; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
ExtruderRear FixedDatum	reference identity must resolve to canonical BOM/drawing; drawing tolerance=barrel bore Ø34.10 +0.05/0; 2×Ø6.6 rail holes; datum face flatness 0.10; bore axis perpendicularity	exports/final/manufacturing/RFQ/ExtruderRearFixedDatum.pdf	docs/build_manual_ko.typ \$ 조립-체결 schedule	NONE	active_part_set reference; canonical=EX-MT-01; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv

Part ID	핵심 인터페이스	도면	조립 단계	필웨어 의존	비고
	0.10/54; hole position ± 0.10 ; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report				
Extruder SupportRail Rear	reference identity must resolve to canonical BOM/drawing; drawing tolerance=length 390.0 ± 0.5 mm; end squareness 0.3; front/rear support height difference ≤ 0.20 mm; axis parallelism $\leq 0.20/390$; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/ExtruderSupportRailRear.pdf	docs/build_manual_ko.typ § 조립·체결 schedule	NONE	active_part_set reference; canonical=EX-MT-04; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FB-01	screen/bin clearance and service withdrawal	docs/drawings/v0.8/FD-002_recirculation_screen.svg	docs/build_manual_ko.typ § Hopper와 cutter	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; real 1.4 mm sheet channels and matching panel corner reliefs
FD-BIN-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=185 x175 x115 outside; PPR-C03 corner reliefs control; no inward burr/dead pocket; removable without cutter disassembly; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-BIN-01.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-HOP-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=OD156 x straight145 + cone55; outlet Ø32; wall2.0; leak test; lid gasket limits moisture ingress; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-HOP-01.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-MET-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=OD29/ID25.00 +0.05/0 x105; flanges Ø44 x3; 4xØ4.5 PCD36 each end; auger radial clearance 0.20-0.25; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-MET-01.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-MET-02	sealed flake path; feeder-to-barrel interface; drawing tolerance=OD24.60 -0.05/0 x105; root Ø10; bore Ø8.2 +0.10/0; pitch18; deburr/polish Ra ≤ 3.2 ; Gate-2 sets mass/rev; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-MET-02.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-MET-03	sealed flake path; feeder-to-barrel interface; drawing tolerance=Ø8 h8 x300; paddle sweep Ø50/100/120 at Z183/228/273; straightness 0.10; 2.2 N·m design torque SF ≥ 2 ; removable drive coupling after donor measurement; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-MET-03.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-TRN-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=centreline length 105.46; OD32, ID28; clock after dry assembly; both sockets ≥ 8 engagement; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-TRN-01.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FH-01	sealed flake path; feeder feedback and cleanability	docs/drawings/v0.8/FD-003_positive_feeder.svg	docs/build_manual_ko.typ § Dry feed와 extruder	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; not a primary dryer; final v0.8 single-path auger architecture; 0.20-0.25 mm radial running clearance; physical mass-per-revolution remains unverified
FH-02	sealed flake path; feeder feedback and cleanability	docs/drawings/v0.8/FD-003_positive_feeder.svg	docs/build_manual_ko.typ § Dry feed와 extruder	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=RECEIPT_TEST_HOLD; quantity_unit=each; external predry not replaced; one-element power/equilibrium test sets final 4-8 quantity; vendor pinned: AliExpress 24 V insulated aluminum-shell PTC 35x21x5 class
FH-03	sealed flake path; feeder feedback and cleanability	docs/drawings/v0.8/FD-003_positive_feeder.svg	docs/build_manual_ko.typ § Dry feed와 extruder	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=USER_APPROVAL_REQUIRED; quantity_unit=each; D44 PWM D42 direction D46 enable D47 fault A7 tach assigned; 5 A branch design envelope; exact donor label ratio voltage current shaft and sensor level required before energization
FM-AX-01	strand alignment and bearing/shaft fit; drawing tolerance=Ø8 h6 x80; TIR0.03; two metal collars; driven spindle interface remains donor-specific; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-AX-01.pdf	docs/build_manual_ko.typ § Cooling, gauge, puller	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-GA-01	strand alignment and bearing/shaft fit; drawing tolerance=Ø5 h6 x30; two E-clips/collars outside PPR-C08; bearing inner-ring clamp must not preload outer rings; no printed axle; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-GA-01.pdf	docs/build_manual_ko.typ § Cooling, gauge, puller	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-GR-01	strand alignment and bearing/shaft fit; drawing tolerance=OD36 x20; 2x Ø16 H7 x5.1-deep 625 seats; Ø12 through relief; seat shoulders square 0.05; groove-free polished surface Ra ≤ 1.6 ; roller only after puller; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-GR-01.pdf	docs/build_manual_ko.typ § Cooling, gauge, puller	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-PL-01	strand alignment and bearing/shaft fit; drawing tolerance=100 x10 x40; 2xØ8.2 roller axes 40.00 apart; 4xØ4.5 guard mounts; matched pair axis position ± 0.05 ; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-PL-01.pdf	docs/build_manual_ko.typ § Cooling, gauge, puller	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-RL-01	strand alignment and bearing/shaft fit; drawing tolerance=finished OD40 x60; bore Ø8.2; TIR ≤ 0.05 ; Shore	exports/final/manufacturing/RFQ/FM-RL-01.pdf	docs/build_manual_ko.typ § Cooling, gauge, puller	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
	A 50-70 sleeve; matched OD within 0.05; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report				user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FR-01	profile joint squareness and table load path	docs/drawings/v0.8/FR-001_frame.svg	docs/build_manual_ko.typ § Frame과 module 배치	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=project lab; cash_class=REUSE_PENDING; status=UNVERIFIED; quantity_unit=each; revision safety-orchestration-closure-v0.6.1; geometry unchanged from 60ccd92fe9a7df35b550a2a57649b1263da09d10; 14.668 m total; local 2040 shredder rails; exports/fabrication/frame_cut_list.csv
GD-01	moving/hot hazard reach protection and service interlock	docs/drawings/v0.8/GD-001_guards_panels.svg	docs/build_manual_ko.typ § 조립-체결 schedule	hardware safety function; firmware monitoring only; never sole protection	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=lot; reach and fragment coupon pending
HP-01	anti-reach opening and removable hopper interface	docs/drawings/v0.8/FD-001_hopper.svg	docs/build_manual_ko.typ § Hopper와 cutter	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; 4.5 L usable; fabrication STEP/DXF/note under exports/fabrication
IN-HOP-01	anti-reach opening and removable hopper interface; drawing tolerance=OD200 x straight150 + cone60; outlet Ø40; wall 2.0; lid rail datum flatness 0.5; leak-free dry-flake seams; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/IN-HOP-01.pdf	docs/build_manual_ko.typ § Hopper와 cutter	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
PL-01	roller alignment, pinch guard and tach calibration	docs/drawings/v0.8/FM-002_gauge_puller.svg	docs/build_manual_ko.typ § Cooling, gauge, puller	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=UNVERIFIED; quantity_unit=each; closed inner speed PI; roller diameter and pulses/rev require CRC calibration; matched plates/rollers and metal spindles
PPR-C01	printed interface per dimension sheet; no structural hot/high-load path; mating=metal hopper rails and lid-interlock flag; tolerance=0.35 mm slide	exports/print/PPR-C01/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C02	printed interface per dimension sheet; no structural hot/high-load path; mating=hopper and metal cutter chamber; tolerance=0.40 mm flake path	exports/print/PPR-C02/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C03	printed interface per dimension sheet; no structural hot/high-load path; mating=1 mm sheet bin and screen rails; tolerance=0.30 mm sheet slot	exports/print/PPR-C03/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C04	printed interface per dimension sheet; no structural hot/high-load path; mating=metal screen; tolerance=0.25 mm	exports/print/PPR-C04/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C05	printed interface per dimension sheet; no structural hot/high-load path; mating=80 mm fan and next duct; tolerance=0.30 mm flange registration	exports/print/PPR-C05/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C06	printed interface per dimension sheet; no structural hot/high-load path; mating=LED/photodiode cross frame and opposite half; tolerance=0.20 mm optical slit	exports/print/PPR-C06/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C07	printed interface per dimension sheet; no structural hot/high-load path; mating=metal puller plate; tolerance=0.40 mm guard gap	exports/print/PPR-C07/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C08	printed interface per dimension sheet; no structural hot/high-load path; mating=FM-GA-01 fixed Ø5 axle and profile; 625 bearings are seated in FM-GR-01; tolerance=Ø5.2 +0.20/0 printed/reamed axle clearance	exports/print/PPR-C08/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C09	printed interface per dimension sheet; no structural hot/high-load path; mating=12 mm metal spindle and metal collar; tolerance=0.30 mm spool core	exports/print/PPR-C09/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C10	printed interface per dimension sheet; no structural hot/high-load path; mating=donor rods and GT2 belt; tolerance=0.20 mm after ream	exports/print/PPR-C10/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C11	printed interface per dimension sheet; no structural hot/high-load path; mating=metal control panel; tolerance=0.25 mm TFT	exports/print/PPR-C11/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C12	printed interface per dimension sheet; no structural hot/high-load path; mating=20 mm profile and fixed 18x18 cable duct; tolerance=18.6 mm cavity; 0.30 mm/side	exports/print/PPR-C12/dimension_sheet.svg	docs/build_manual_ko.typ § Print package	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-EXTRUDER-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/EX-001_extruder_assembly.svg	docs/build_manual_ko.typ § 조립-체결 schedule	NONE	active_part_set reference; canonical=PPR-EXTRUDER-ASM; exclude from procurement roll-up
PPR-FEEDER-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/FD-003_positive_feeder.svg	docs/build_manual_ko.typ § 조립-체결 schedule	NONE	active_part_set reference; canonical=PPR-FEEDER-ASM; exclude from procurement roll-up
PPR-FORMING-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/FM-001_cooling_strand_path.svg	docs/build_manual_ko.typ § 조립-체결 schedule	NONE	active_part_set reference; canonical=PPR-FORMING-ASM; exclude from procurement roll-up
PPR-FRAME-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/FR-001_frame.svg	docs/build_manual_ko.typ § 조립-체결 schedule	NONE	active_part_set reference; canonical=PPR-FRAME-ASM; exclude from procurement roll-up
PPR-FULL-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/ASM-001_full_assembly.svg	docs/build_manual_ko.typ § 조립-체결 schedule	NONE	active_part_set reference; canonical=PPR-FULL-ASM; exclude from procurement roll-up
PPR-SHREDDER-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/build_manual_ko.typ § 조립-체결 schedule	NONE	active_part_set reference; canonical=PPR-SHREDDER-ASM; exclude from procurement roll-up
SF-01	hardwired E-stop/interlock/thermal chain independent of firmware	docs/drawings/v0.8/EL-001_electrical_enclosure.svg	docs/build_manual_ko.typ § Control과 UI	hardware safety function; firmware monitoring only; never sole protection	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=MIXED; status=PHYSICAL_GATE; quantity_unit=each; software independent
SF-02	hardwired E-stop/interlock/thermal chain independent of firmware	docs/drawings/v0.8/EL-001_electrical_enclosure.svg	docs/build_manual_ko.typ § Control과 UI	hardware safety function; firmware monitoring only; never sole protection	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=MIXED; status=PHYSICAL_GATE; quantity_unit=each; hardware branch inhibit
SF-03	hardwired E-stop/interlock/thermal chain independent of firmware	docs/drawings/v0.8/EL-001_electrical_enclosure.svg	docs/build_manual_ko.typ § Control과 UI	hardware safety function; firmware monitoring only; never sole protection	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=MIXED; status=PHYSICAL_GATE; quantity_unit=lot; not optional; vendor path: domestic parts malls SF139E-class thermal fuse plus branch fuse holder; exact rating per heater RFQ receipt contract
SH-01	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/build_manual_ko.typ § Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=COUPON_HOLD; quantity_unit=each; common PLA/PET path
SH-02	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/build_manual_ko.typ § Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=COUPON_HOLD; quantity_unit=each; metal shim clearance
SH-03	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/build_manual_ko.typ § Hopper와 cutter	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=project lab/RFQ; cash_class=REUSE_PENDING; status=DONOR_HOLD; quantity_unit=each; 12T:30T

Part ID	핵심 인터페이스	도면	조립 단계	필웨어 의존	비고
					cutter 20-40 rpm 14 N.m continuous/24 N.m 3 s peak; exact model label state shaft current temperature and Gate-1 required
SH-04	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/build_manual_ko.typ § Hopper와 cutter	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=COUPON_HOLD; quantity_unit=each; 22 N.m is cutter-equivalent; motor-side relief is 17.25/12.94/10.35 N.m by ratio
SH-05	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/build_manual_ko.typ § Hopper와 cutter	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=buy; cash_class=CASH; status=RECEIPT_HOLD; quantity_unit=each; calibrated 18 N.m jam trip plus 20 A branch fuse; bench-calibrate before cutter; vendor pinned: icbanq BTS7960 module P014373686 (4
SP-01	spindle/bearing fit; dancer/traverse envelope	docs/drawings/v0.8/SP-001_spooler_traverse.svg	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; 1 kg spool; A15 spool tach and A5/A6 limits; radius remains geometry/wound-length estimate with uncertainty
SP-AX-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=Ø8 h6 x28; metal collars; one pivot and one end roller axle; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-AX-01.pdf	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-BP-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=105 x5 x60; bearing centre X30, Ø28.2; 4xØ5.5 bearing block + 2xØ5.5 profile tab; matched axis position ±0.05; bearing outer ring retained by metal washer/clip; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-BP-01.pdf	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-DA-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=100 mm pivot centres; 12 mm arm; 2xØ8.2; edge R2; full -25..+25deg motion; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-DA-01.pdf	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-DS-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=36 x8 x80; pivot Ø8.2 at X18/Z45; 2xØ5.5 foot mounts; metal support carries spring/tension load; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-DS-01.pdf	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-MM-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=101 x6 x52; motor centre X26, Ø24; 4xØ4.5 at 31 mm square; 2xØ5.5 profile tab; actual donor shaft and body measurement required before coupling release; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-MM-01.pdf	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-RL-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=OD20 x20; bore Ø8.2; free rotation under 0.2-1.0 N filament tension; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-RL-01.pdf	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-SH-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=Ø12 h6 x143; straightness0.05; two 6001 bearings; axial collars carry spool load; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-SH-01.pdf	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-TR-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=5 x50 x40; two Ø8.2 at 25 mm spacing; 2xØ5.5 mount; matched pair and rod parallelism <=0.10/160; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-TR-01.pdf	docs/build_manual_ko.typ § Guide, dancer, traverse, spool	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
TH-BH-01	heater fit, insulation, branch fuse and independent thermal cutoff; drawing tolerance=24 VDC 100 W each; as-clamped ID34.00 +0.10/0; width45 ±0.5; radial build2 nominal; cold resistance 5.76 Ω ±10%; 300 mm fiberglass leads; grounded sheath; PET service 300 C design; no stock Ø35 substitution; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/TH-BH-01.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
TH-DIE-01	heater fit, insulation, branch fuse and independent thermal cutoff	exports/thermal/parts/TH-DIE-01/drawing_notes.md	docs/build_manual_ko.typ § Dry feed와 extruder	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
TH-FUSE-01	heater fit, insulation, branch fuse and independent thermal cutoff	exports/thermal/parts/TH-FUSE-01/drawing_notes.md	docs/build_manual_ko.typ § Dry feed와 extruder	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
TH-PTC-01	heater fit, insulation, branch fuse and independent thermal cutoff; drawing tolerance=120 x55 x3; 4x Ø4.5; flatness0.20; electrically bonded to hopper; PTC-to-metal insulation system supplier-rated; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/TH-PTC-01.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
TH-PTC-02	heater fit, insulation, branch fuse and independent thermal cutoff; drawing tolerance=120 x55 x2; 4x Ø4.5; spring/compliant electrically-insulating pads prevent ceramic point load; PE bond; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/TH-PTC-02.pdf	docs/build_manual_ko.typ § Dry feed와 extruder	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
TH-PTC-EL	heater fit, insulation, branch fuse and independent thermal cutoff	exports/thermal/parts/TH-PTC-EL/drawing_notes.md	docs/build_manual_ko.typ § Dry feed와 extruder	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
TH-TC-01	heater fit, insulation, branch fuse and independent thermal cutoff	exports/thermal/parts/TH-TC-01/drawing_notes.md	docs/build_manual_ko.typ § Dry feed와 extruder	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval

4 일정표 요약

Fastener 22행 · Consumables 5행 · Tools 13행 · Alternatives 9행.

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